



Nurturing Minds For a Better World
VOL. 03 | ISSUE. 01

Sahyadri
TECH X PRESS

REFLECTIONS OF IIT PALAKKAD



AT A GLANCE

Established in 2015, the Indian Institute of Technology Palakkad is dedicated to creating an environment that enables students and faculty to engage in the pursuit of knowledge, dream, think, and innovate, thereby becoming change agents for a better world. Having begun with just 120 students in July 2015, IIT Palakkad has since grown into an institution with nearly a thousand students and a highly accomplished faculty and staff in key positions.

The Institute recognises collective growth through collaboration with industry and other academic institutions as the need of the hour and emphasises blue-sky research and directed research as two essential pillars of technology development. The Institute embraces a vision of becoming a leader in the fundamental research and technological innovation. This vision is embodied in the Institute's motto, 'Nurturing Minds for a Better World'.

Indian Institute Of
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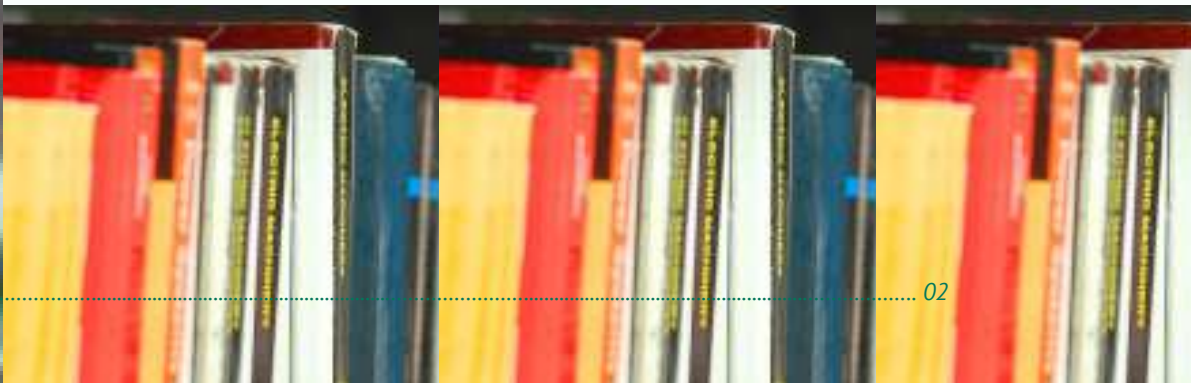
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DIRECTOR'S DESK

It gives me immense pleasure to present this edition of the IIT Palakkad Newsletter, Sahyadri TechXpress, Vol. 03, Issue 01, showcasing some of the significant milestones and accomplishments that have marked the Institute's journey in recent months. As we embark on a new academic year, I extend a warm welcome to our incoming students, who now become part of a vibrant community committed to excellence in education, research, innovation, and nation-building.

The past few months have been particularly eventful for the Institute. We have expanded our academic landscape with the launch of the Department of Materials and Metallurgical Engineering and welcomed a larger undergraduate cohort this year. We also marked a significant milestone through our collaboration with the Ministry of Housing and Urban Affairs and the establishment of the Centre for Digital Construction and Management (CDCM), reinforcing our commitment to digital transformation and capacity building. The 2nd International Conference on Non-Sewered Sanitation Systems (ICNSS 2026), organised by our Global Sanitation Centre of Excellence (GSCoE), further strengthened IIT Palakkad's global engagement and collaborative research ecosystem.

I hope this edition provides an insightful glimpse into the many achievements and activities of our Institute. I wish our new students a rewarding and enriching journey at IIT Palakkad, and I look forward to another year of learning, innovation, and collective success.



Prof. A Seshadri Sekhar, Director, IIT Palakkad



"Excellence is built through curiosity, collaboration, and a shared commitment to creating meaningful impact. At IIT Palakkad, we continue to nurture an ecosystem where ideas translate into solutions for society."

EDITORIAL BOARD

Editors: Ms. Devika Sujith, Ms. Sreeja OK

Guest Editor: Prof. V Haridas

Designs and Content: Ms. Sreeja OK, Ms. Devika Sujith

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The background of the page is a photograph of the IIT Palakkad campus. In the foreground, there is a large, multi-story building with a red-tiled roof and a checkered facade. Behind it, other campus buildings are visible, and in the far background, there are green hills and mountains under a clear sky.

The Department of Materials and Metallurgical Engineering at IIT Palakkad is dedicated to advancing education, research, and innovation in materials science and engineering—an area that forms the foundation of virtually every technological advancement. From next-generation semiconductor materials powering AI hardware beyond conventional silicon to advanced materials for infrastructure, transportation, healthcare, and clean energy, the discipline plays a transformative role in addressing global technological challenges. The department provides a strong academic framework that spans the entire spectrum of materials engineering, encompassing the understanding of atomic-scale phenomena and microstructural behaviour, as well as the design, processing, characterisation, and development of advanced materials for real-world applications. Through a rigorous curriculum supported by modern laboratories, interdisciplinary learning, research opportunities, and industry engagement, students gain a comprehensive understanding of both fundamental principles and emerging technologies.

DEPARTMENT OF MATERIALS AND METALLURGICAL ENGINEERING

As the field continues to evolve from conventional trial-and-error methods to science-driven and data-enabled materials design, the department prepares students to contribute to frontier areas such as semiconductors, batteries, hydrogen technologies, solar cells, biomaterials, additive manufacturing, and sustainable materials engineering. Research-led teaching, collaborations with leading academic institutions, and partnerships with industry ensure that learning remains relevant to societal and industrial needs. By fostering innovation, critical thinking, and multidisciplinary problem-solving, the department aims to develop graduates who are technically proficient, research-oriented, and socially responsible, and who are ready to pursue impactful careers in core industries, advanced research, higher education, and entrepreneurship while contributing to India's technological and manufacturing aspirations.

“COMMITTED TO CREATING SUSTAINABLE SOLUTIONS FOR THE SOCIETY”

HOD’S MESSAGE



Prof. Mayank Tiwari
Head-In-Charge
Department of MME

“The Department of Materials and Metallurgical Engineering at IIT Palakkad aspires to build a vibrant ecosystem of learning, research, and innovation in one of the most significant disciplines shaping the future of technology. Our focus is on providing students with a strong foundation in materials science while exposing them to emerging domains such as nanomaterials, biomaterials, energy materials, and sustainable materials engineering.

By integrating rigorous academics with hands-on laboratory experience and interdisciplinary research, we seek to equip students to address complex engineering challenges through innovation. We also envision strong collaborations with global and national industries and research organisations, particularly across Kerala and Coimbatore, to facilitate internships, joint research, and technology-driven problem-solving. Through these efforts, we aim to nurture engineers and researchers who are technically competent, innovative in outlook, and committed to creating sustainable solutions for society.”

Prof. Mayank Tiwari
Head-In-Charge

HIGH-PERFORMANCE SODIUM-ION BATTERIES USING FLEXIBLE 3D POLYIMIDE CATHODES

The rapid expansion of renewable energy and portable electronics has increased global demand for efficient, affordable, and sustainable energy storage solutions. While lithium-ion batteries currently lead the market, concerns about limited lithium supplies and rising costs have spurred the search for alternatives.

In this regard, sodium-ion batteries (SIBs) are promising options because sodium is abundant, inexpensive, and widely available. Organic electrode materials have attracted considerable interest for SIB applications due to their lightweight, eco-friendly nature and the abundance of their constituent elements.

However, many organic cathodes face challenges, including poor structural stability and high solubility in electrolytes during repeated charge-discharge cycles, which hinder their durability.

To address these issues, Dr. Shanmugaraju Sankarasekaran's research group in the Department of Chemistry at IIT Palakkad has developed a flexible, 3D cross-linked polyimide cathode material, NDI-TREN, for high-performance sodium-ion batteries. This polymer was synthesized via a straightforward polycondensation reaction between 1,4,5,8-naphthalenetetracarboxylic dianhydride (NTCDA) and tris(2-aminoethyl)amine (TREN), as shown in Scheme 1. The porous 3D structure of NDI-TREN facilitates rapid sodium-ion diffusion, while the electroactive carbonyl groups enable highly reversible redox reactions during battery operation. Notably, the electrode was fabricated using an environmentally friendly, water-based process with a carboxymethyl cellulose (CMC) binder, thereby avoiding toxic organic solvents. Electrochemical analysis showed that NDI-TREN provides a high specific capacity of 130 mAh g^{-1} , along with excellent rate capability and strong long-term cycling performance. Notably, it retained nearly 80% of its capacity after 8,300 cycles at 500 mA g^{-1} , with near-perfect coulombic efficiency. Even at higher current densities, the polymer demonstrated excellent structural stability and highly reversible electrochemical behavior (Figure 1).



Scheme 1: One-pot synthesis of polyimide NDI-TREN from monomers NTCDA and TREN.

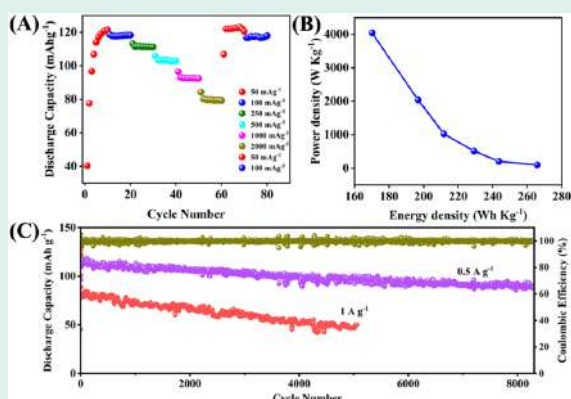


Figure 1 : A) Galvanostatic charge and discharge profiles
B) Rate performance
C) Long-term cycling performance of the NDI-TREN.

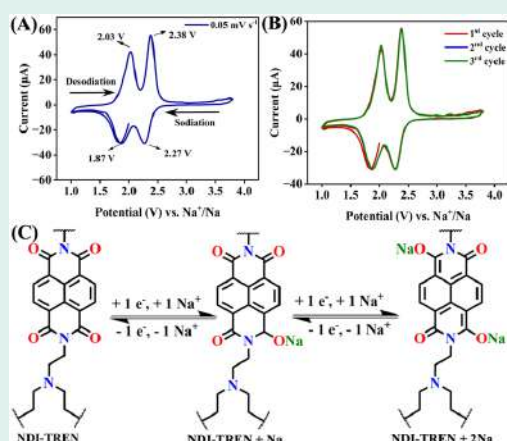


Figure 2: Electrochemical redox mechanism of NDI-TREN

Detailed computational studies and post-mortem analyses validated the reversible sodium-ion storage mechanism and the polymer's outstanding structural stability over extended cycling (Figure 2). This research highlights the significant potential of sustainable organic polymer cathodes for future sodium-ion batteries and environmentally friendly, large-scale energy storage solutions.

This work was conducted by Ph.D. students Mr. A. Vamshi Krishna and Ms. Dimple Deori (IIT Madras) under the supervision of Dr. Shanmugaraju Sankarasekaran and Dr. Thangavelu Palaniselvam (IIT Madras), respectively. The computational studies were carried out by Dr. Deivasigamani Umadevi (DST Scientist) at IIT Palakkad. An Indian patent (Patent No. 588970) has been granted for this work, and the findings have been submitted for publication. Also this work has been accepted for publication in the journal "Energy Technology".



Mr. A Vamshi Krishna

Research Scholar
Department of Chemistry
IIT Palakkad

Dr. Shanmugaraju S

Associate Professor
Department of Chemistry
IIT Palakkad



Ms. Dimple Deori

Research Scholar
IIT Madras

Dr. Thangavelu Palaniselvam

Collaborator
IIT Madras



Dr. Deivasigamani Umadevi

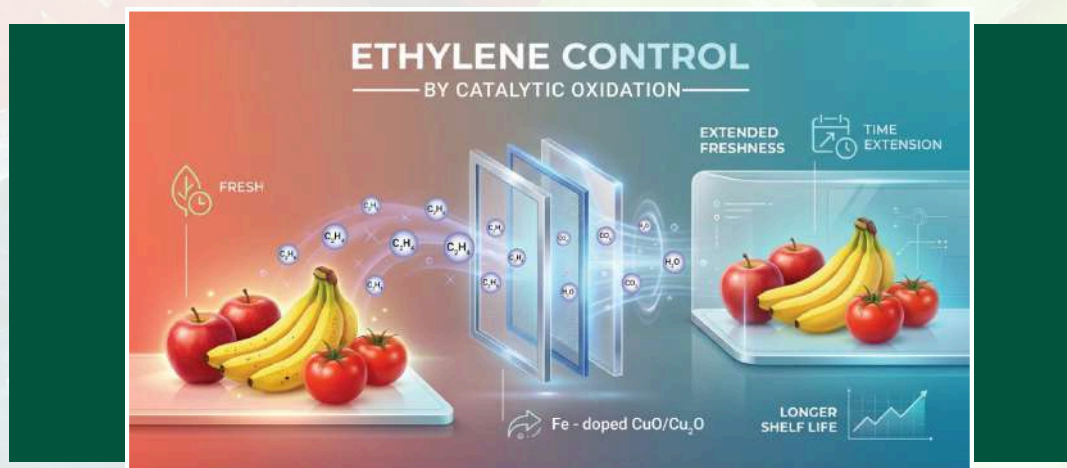
DST Scientist
IIT Palakkad



IIT PALAKKAD RESEARCHERS DEVELOP NOVEL LOW-TEMPERATURE CATALYST FOR POST-HARVEST CROP PRESERVATION

India's vast geographical diversity necessitates the transportation and storage of agricultural produce over extensive distances. Ensuring the timely distribution and preservation of crops is therefore critical to maintaining food quality and minimizing economic losses. Post-harvest spoilage remains a persistent challenge, particularly for climacteric fruits such as bananas, which continue to ripen after harvest due to the production of ethylene, a naturally occurring plant hormone.

Ethylene plays a crucial role in the ripening process; however, its excessive accumulation during storage can accelerate ripening and lead to the rapid deterioration of fruits. This not only diminishes their nutritional value but also adversely affects their commercial quality, contributing significantly to food wastage and financial losses across the agricultural supply chain.



One sustainable approach to addressing this challenge involves the use of catalytic converters within storage containers to remove excess ethylene from the environment. While similar in principle to catalytic converters used in internal combustion engine exhaust systems, such technologies for agricultural applications must operate efficiently at the low temperatures typically maintained in cold-storage facilities.

In a significant technological advancement, researchers from the Department of Chemistry at IIT Palakkad have developed a novel transition-metal-based catalyst capable of operating at temperatures as low as 5°C.

The innovation, recently patented by Dr. Dinesh Jagadeesan and Ms. M. Roshni, offers a practical solution for ethylene management under cold-storage conditions, thereby helping to extend the shelf life of perishable produce.



Dr. Dinesh Jagadeesan



Ms. M. Roshni

A notable feature of the catalyst is its ability to function without the use of expensive precious metals such as platinum, gold, palladium, and silver, which are conventionally employed in catalytic technologies. This makes the invention both economically viable and scalable for widespread agricultural deployment.

‘The technology represents a promising contribution to sustainable post-harvest management practices. By enabling effective ethylene control under low-temperature conditions, it has the potential to reduce food wastage, preserve crop quality, and enhance economic returns for farmers and stakeholders across the agricultural value chain.’



ISHAN AGRAWAL

Scientific Officer/D (Executive Engineer)
Nuclear Infrastructure Projects
Nuclear Power Corporation of India Limited
(NPCIL)

Civil Engineering, Class of 2019

Hi, I'm Ishan Agrawal, a Civil Engineering graduate from IIT Palakkad, from the 2019 batch. After graduating, I joined the Nuclear Power Corporation of India Limited (NPCIL), where I currently work as an Executive Engineer/Scientific Officer on nuclear infrastructure projects. My work involves planning, design review, quality assurance, and execution of large civil engineering projects. Beyond my profession, I have a keen interest in personal finance, investing, and continuous learning

What role did IIT Palakkad play in shaping your career?

IIT Palakkad gave me much more than a degree - it taught me how to think. Prof. Mathews said on the first day that I am not here to cover the syllabus, I am here to uncover the syllabus.

Being part of one of the institute's first batches meant learning to adapt, take initiative, and build things from the ground up. The rigorous academics strengthened my engineering fundamentals, while the close interaction with faculty and peers encouraged curiosity and independent problem-solving. The exposure I received to civil design software was unmatched which still gives me an edge over my colleagues.

The confidence, discipline, and analytical mindset I developed at IIT Palakkad played a significant role in helping me secure a good GATE rank and eventually build a career with NPCIL.

What is your message to your juniors?

Don't measure your success only by placements or salaries. Focus on building strong fundamentals, developing problem-solving skills, and staying curious. The opportunities that come your way will keep changing, but your ability to learn will always stay valuable. Make the most of your time at IIT Palakkad - participate in projects, interact with professors, build meaningful friendships, and don't hesitate to step outside your comfort zone. Finally, remember that careers are a marathon, not a sprint. Stay consistent, stay humble, and keep learning.

I graduated from IIT Palakkad in 2020 with a B.Tech. in Computer Science. My work has focused on robotics and autonomous systems. I enjoy working on drones, AMRs, and robotic arm. I've won recognition at DRDO Dare-to-Dream (felicitated by the Defence Minister twice) and qualified for the MBZIRC, SIH, and OpenCV Spatial AI contests. I'm currently a software engineer at Siemens.

What role did IIT Palakkad play in shaping your career?

IIT Palakkad gave me my first real exposure to robotics. I headed the institute's Robotics Club and mentored teams for Hackathons, and Inter-IITs. My bachelor's thesis on SLAM, swarms, and reinforcement learning for multi-robot systems set the direction for everything I've worked on since. The institute's small, close-knit environment in its early years also meant I got to learn directly from professors and peers across disciplines, which broadened how I think about problems.



RAJENDRA SINGH

Software Engineer (R&D) at Siemens
Software

Computer Science Engineering
Class of 2019

What is your message to your juniors?

"Don't wait for a course to hand you a project and go build something, even if it's as small as 3D-printed robot arm in a lab. The clubs, competitions, and open-source communities will teach you more than lectures ever will. And start early: the earlier you get your hands dirty, the more room you'll have to fail, learn, and try again before it actually counts."



GC3

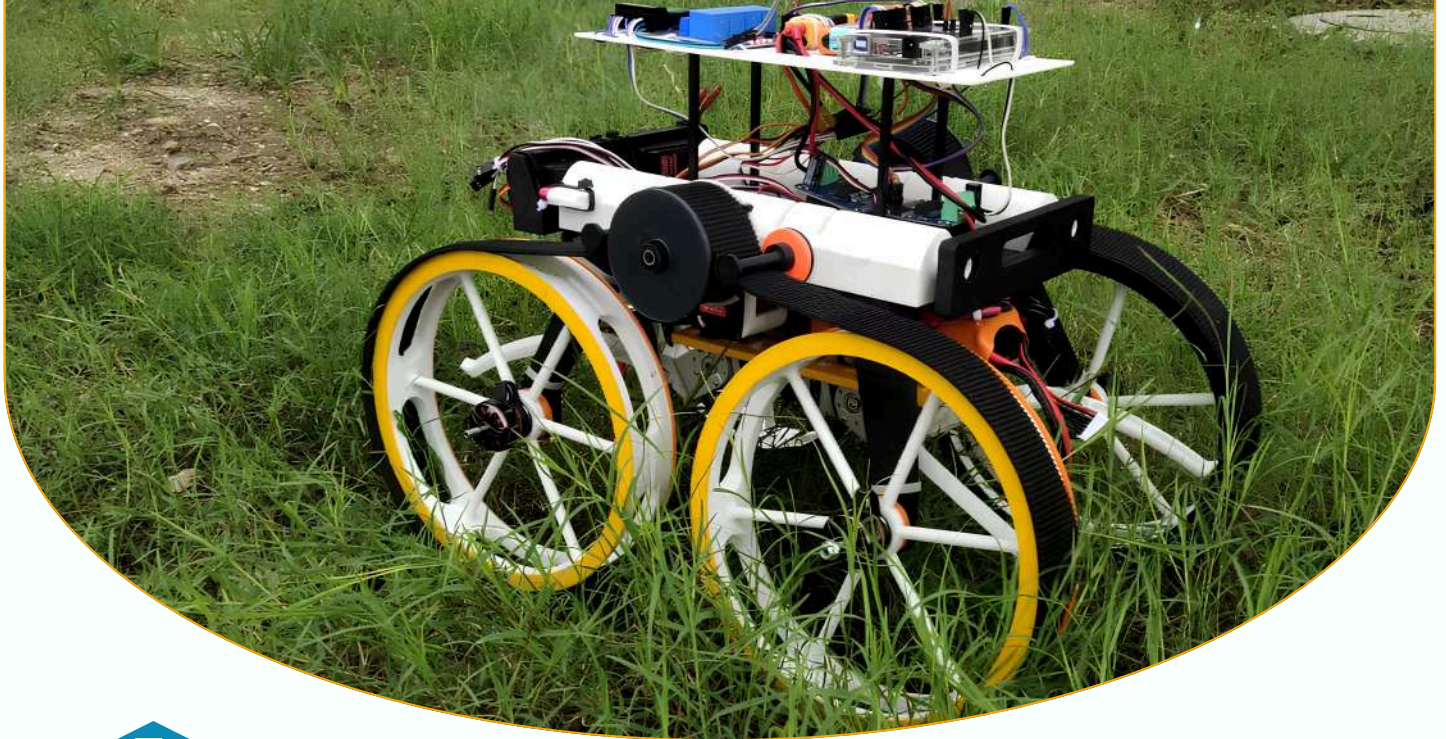
GENERAL CHAMPIONSHIP 2026

General Championship 2026 was a grand celebration of talent and sportsmanship! The event recognized the outstanding performances of the Wild Wolves, Stalwart Rhinos, Mighty Lions, Stealthy Panthers, Ferocious Tigers, Majestic Eagles, and Savage Sharks.

IIT Palakkad was honoured to have Mr. Joseph Abraham, recipient of the Arjuna Award and a gold medallist at the Asian Games, as the chief guest







A DUAL-MODE UNMANNED VEHICLE RECONFIGURATION SYSTEM

This invention discloses a novel reconfiguration system for unmanned vehicles designed to operate across multiple environmental domains through structural adaptability. The system utilizes a specialized mechanism that enables the vehicle to transition between distinct operational modes, typically optimized for terrestrial and aquatic (or aerial) navigation, thereby overcoming the limitations of fixed-frame robotic platforms.

By integrating an intelligent control framework with a transformable architecture, the system ensures high maneuverability, stability, and energy efficiency during the transition phase. This innovation significantly enhances the utility of autonomous systems in complex, hybrid environments, such as disaster zones, coastal surveillance, and industrial inspection, offering a versatile solution that eliminates the need for multiple environment-specific robots.

INVENTORS



Prof. Santhakumar Mohan



Mr. Amar Hamdan Thayyil



Mr. Labeeb Nassar

CONVOCATION 2026

The 8th Convocation Ceremony for the graduating batch of 2026 was held on 18 July at the Institute's Agora Auditorium. Dr. Tessy Thomas, Former Director General (Aeronautical Systems) at the Defence Research and Development Organisation (DRDO), popularly known as the 'Missile Woman of India,' graced the occasion as the chief guest. Dr. Tessy Thomas also presented the Institute Medals to students in recognition of their outstanding academic performance. Gayathri S. received the Institute Medal for B.Tech., while Manal Jaleel Poovathingal was awarded the Institute Medal for M.Tech. In addition, students who secured the highest CGPA in their respective departments were honoured for their academic excellence, and awards were presented to students for the Best Projects across various programmes.

A total of 305 students graduated during the ceremony, with degrees awarded across various programmes, including 149 B.Tech., 67 M.Tech., 46 M.Sc., 30 Ph.D., and 13 M.S. (by Research) degrees.



EVENTS AND COLLABORATIONS



Inauguration of the 500 MHz Nuclear Magnetic Resonance



Delegation from Southern Cross University (SCU), Australia



World Environment Day Celebration 2026



SATVA 2026



Institute Dashboard launch



RECCAP 2026



SAMAVESH 2.0



Independence Day Celebration 2026



ICNSS 2026



MoU with Bureau of Indian Standards



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REACH US



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भारतीय प्रौद्योगिकी संस्थान पालक्काड
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